Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Face verification and face recognition are the two most common names given to the task of comparing a new picture against one person's face. True/False?

1 / 1 point

- (●) False
- ( ) True

Correct
Correct. This is the description of face verification, but not of face recognition.

2. Why do we learn a function $d(img1, img2)$ for face verification? (Select all that apply.)

1 / 1 point

- [ ] This allows us to learn to predict a person's identity using a softmax output unit, where the number of classes equals the number of persons in the database plus 1 (for the final "not in database" class).
- [x] We need to solve a one-shot learning problem.

Correct
This is true as explained in the lecture.

- [ ] Given how few images we have per person, we need to apply transfer learning.
- [x] This allows us to learn to recognize a new person given just a single image of that person.

Correct
Yes.

3. In order to train the parameters of a face recognition system, it would be reasonable to use a training set comprising 100,000 pictures of 100,000 different persons.

1 / 1 point

- ( ) True
- (●) False

Correct
Correct, to train a network using the triplet loss you need several pictures of the same person.

4. In the triplet loss:

1 / 1 point

$$\max\left(\|f(A)-f(P)\|^2 - \|f(A)-f(N)\|^2 + \alpha, 0\right)$$

Which of the following are true about the triplet loss? Choose all that apply.

- [ ] A the anchor image is a hyperparameter of the Siamese network.
- [x] We want that $\|f(A)-f(P)\|^2 < \|f(A)-f(N)\|^2$ so the negative images are further away from the anchor than the positive images.

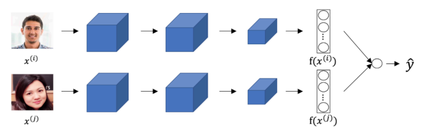
Correct
Correct. Being a positive image the encoding of P should be close to the encoding of A.

- [ ] α is a trainable parameter of the Siamese network.
- [x] $f(A)$ represents the encoding of the Anchor.

Correct
Correct. f represents the network that is in charge of creating the encoding of the images, and A represents the anchor image.

5. Consider the following Siamese network architecture:

1 / 1 point



Which of the following do you agree with the most?

- ( ) Although we depict two neural networks and two images, the two images are combined in a single volume and pass through a single neural network.
- (●) The upper and lower neural networks depicted have exactly the same parameters, but the outputs are computed independently for each image.
- ( ) The two neural networks depicted in the image have the same architecture, but they might have different parameters.
- ( ) This depicts two *different* neural networks with different architectures, although we use the same drawing.

Correct
Correct. Both neural networks share the same weights, and each image passes through the neural network in an independent manner.

6. Our intuition about the layers of a neural network tells us that units that respond more to complex features are more likely to be in deeper layers. True/False?

1 / 1 point

- ( ) False
- (●) True

Correct
Correct. Neurons that understand more complex shapes are more likely to be in deeper layers of a neural network.

7. Neural style transfer is trained as a supervised learning task in which the goal is to input two images (x), and train a network to output a new, synthesized image (y).

1 / 1 point

- ( ) True
- (●) False

Correct
Yes, Neural style transfer is about training the pixels of an image to make it look artistic, it is not learning any parameters.

8. In neural style transfer the content loss J_{cont} is computed as:

1 / 1 point

$$J_{cont}(G, C) = \left\|a^{[l](C)} - a^{[l](G)}\right\|^2$$

Where $a^{[l](k)}$ is the activation of the l-th layer of a ConvNet trained for classification. We choose l to be a very high value to use compared to the more abstract activation of each image. True/False?

- ( ) True
- (●) False

Correct
Correct. We don't use a very deep layer since this will only compare if the two images belong to the same category.

9. In neural style transfer, we can't use gradient descent since there are no trainable parameters. True/False?

1 / 1 point

- ( ) True
- (●) False

Correct
Correct. We use gradient descent on the cost function $J(G)$ and we update the pixel values of the generated image G.

10. You are working with 3D data. The input "image" has size $64 \times 64 \times 64 \times 3$, if you apply a convolutional layer with 16 filters of size $4 \times 4 \times 4$, zero padding and stride 2. What is the size of the output volume?

1 / 1 point

- ( ) $64 \times 64 \times 64 \times 3$.
- ( ) $31 \times 31 \times 31 \times 3$.
- (●) $31 \times 31 \times 31 \times 16$.
- ( ) $61 \times 61 \times 61 \times 14$.

Correct
Correct, we can use the formula $\left\lfloor \frac{n^{[l-1]} - f + 2 \times p}{s} \right\rfloor + 1 = n^{[l]}$ to the three first dimensions.